A.3 Brier Score

We use the standard prior score of

$$B = \frac{1}{N}\sum_{i}^{N}(p_i - c_i)^2 \tag{A.3}$$

where p_i = P(True) for any given sample (i.e. the model's self-evaluated probability that it's sample was correct), and c_i is 0 if the sample is incorrect, and 1 if it is correct. In contexts where many samples have been generated at unit temperature, we binarize the ground-truth P(IK) via $\mathbb{1}\{\text{Ground-Truth P(IK)} > 0.5\}$.

A.4 BIG Bench Default Format

By default in BIG Bench [Srivastava et al., 2022] questions are posed in this way:

```
Question: Who was the first president of the United States?
 choice: Barack Obama
 choice: George Washington
 choice: Michael Jackson
Answer:
```

Here models are required to re-write the complete responses. We find that calibration is poorer in this format, compared to a format with lettered choices, and where models need only write that letter.

A.5 Prompt for Self-Evaluation

We used this prompt for self-evaluation of P(True), when not taking a few-shot approach (which performs best, as it improves significantly on calibration). This was simply the first prompt we tried, and it was not otherwise optimized. It contains six examples, with some similarity to arithmetic, trivia, and fill-in-the-blank completions. This version contains extra "brainstormed" examples; we also used an otherwise identical version without these examples for the types of evaluation discussed in section 4.1:

```
Question: Who was the third president of the United States?
Here are some brainstormed ideas: James Monroe\n Thomas Jefferson\n Jefferson\n
    Thomas Jefferson\n George Washington
Possible Answer: James Monroe
Is the possible answer:
 (A) True
 (B) False
The possible answer is: (B)

Question: Calculate 33 + 4
Here are some brainstormed ideas: 37\n 37\n 40\n 36\n 37
Possible Answer: 37
Is the possible answer:
 (A) True
 (B) False
The possible answer is: (A)

Question: Fill in the blank in the sentence "I went to the grocery and then to
    the pharmacy. I was disappointed that they didn't have any vegetarian
    sausage at the _____."
Here are some brainstormed ideas: grocery\n store\n grocery\n refrigerator\n
    grocery
Possible Answer: grocery
Is the possible answer:
 (A) True
 (B) False
The possible answer is: (A)

Question: Name a celebrated civil rights leader.
Here are some brainstormed ideas: Martin Luther King\n Ghandhi\n Martin Luther
    King\n Barack Obama\n Martin Luther King
Possible Answer: Martin Luther King
```

```
Is the possible answer:
 (A) True
 (B) False
The possible answer is: (A)

Question: Calculate 33 * 849
Here are some brainstormed ideas: 28,347\n 1,490\n 27,488\n 3,409\n 34,561
Possible Answer: 28347
Is the possible answer:
 (A) True
 (B) False
The possible answer is: (B)

Question: Fill in the blank in the sentence "I shot the _____ and it went swish.
     We walked away the winners of that battle!"
Here are some brainstormed ideas: gun\n bullet\n arrow\n basketball\n rifle
Possible Answer: gun
Is the possible answer:
 (A) True
 (B) False
The possible answer is: (B)
```

A.6 Training for P(IK)

In order to train P(IK) classifiers, we created datasets that consisted of (Q, IK / IDK Label) pairs. Given a question Q, such as a TriviaQA question that includes a few-shot prompt, we generated 30 candidate answer samples at $T = 1$. See Appendix A.7 for details on the formatting of Q on each of our tasks. We then scored each candidate answer sample as either correct or incorrect. For each candidate answer, we then create a datapoint (Q, True Label) based on whether or not the model got the answer correct. Thus, for each question, we have 30 datapoints of (Q, True Label). This is done as a simple way of approximating a soft label using many hard labels.

During fine-tuning, we used a batch size of 7680, and a learning rate that was $\frac{1}{3}$ of the original pretraining learning rate. We noticed that using a larger than normal batch size was important for learning because of our noisy soft training labels.

The evaluation datasets are constructed similarly to the above, except during evaluation, we focus on computing ground truth P(IK) scores for each question, instead of many IK/IDK labels for each question. The ground truth P(IK) is simply the fraction of $T = 1$ samples the model gets correct. We then compare this ground truth P(IK) score to the model's predicted P(IK) score on these evaluation questions in many plots throughout the paper.

A.7 Formatting of Questions for P(IK)

TrivaQA: TriviaQA was done 10-shot. For brevity, here is a 2-shot example:

```
Question: Anatomy. Where are the intercostal muscles situated?

Answer: between the ribs
Question: What is the thick watery substance filling the space between
   the lens and the retina of the eye?

Answer:
```

Lambada: Lambada was done 10-shot. For brevity, here is a 2-shot example:

```
Question: Please complete the following passage by filling in the blank:
     John said, "Hi, this is the Telecommunications Exchange for the
    Deaf, and I have Gloria on the phone. She's hearing impaired, so I'm
     calling for her. Is this Alan?"
"Yes."
"OK, please hold."
```

```
John told the deaf person that he had Alan on the line. Then she began
    to type, and John read her message to Alan: "Hi, Alan, this is _____.
     ->

Answer: gloria

Question: Please complete the following passage by filling in the blank:
     He was good looking, but she was not in the mood to be picked up by
     anyone.
"I'm McKenzie," said the man. He reached out his free hand to Ana.
Ana hesitated for a moment, then shook McKenzie's hand and said, "Ana."
"Where are you from, Ana?"
"Look, if you are trying to pick up a girl, I am not it," said _____. ->

Answer:
```

Arithmetic: Arithmetic was done 10-shot. For brevity, here is a 2-shot example:

```
Question: What is 36 plus 32?

Answer: 68
Question: What is 7,781 minus 5,071?

Answer:
```

Python Function Synthesis: Python Function Synthesis was done 0-shot. The context for each Python Synthesis problem is a partial Python file, with a function definition and docstring at the very end. The model is then tasked with filling in the function so that it passes a test.

```
...
<Beginning of file is given as context>
...

def make_random_slice(upper):
   """Return a slice of an array with upper bound upper. Guaranteed to
      have
   at least one element."""
```

GSM8k: GSM8k was done 10-shot. For brevity, we have included a 2-shot example below. Note that the model is prompted to output reasoning as well as the correct answer. However, each candidate answer is scored based on the number given after the `####`.

```
Question: Verna loves to eat fruit. She bought three apples at $1.50
    each, five oranges at $0.80 each, and six peaches at $0.75 each. If
    she gave $20, how much change did she receive?

Answer: three apples cost 3 x $1.50 = $<<3*1.5=4.50>>4.50.
five oranges cost 5 x $0.80 = $<<5*0.8=4>>4.
four peaches cost 6 x $0.75 = $<<6*0.75=4.50>>4.50.
thus, verna paid a total of $4.50 +$4+ $4.50 = $<<4.5+4+4.5=13>>13.
therefore, verna received $20 - $13 = $<<20-13=7>>7.
#### 7
Question: Frederick is making popsicles to sell and to save money he is
    making his own popsicle sticks. He can get 200 sticks from a 2 x 4
    piece of wood and 400 sticks from a 2 x 8 piece of wood. He has $24
    to buy wood for sticks. A 2 x 4 costs $4. A 2 x 8 costs $6. What is
    the most popsicle sticks he can make if he buys the cheapest lumber?

Answer:
```